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Error estimates for the finite element method of the Navier-Stokes-Poisson-Nernst-Planck equations ⋆

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ABSTRACT

In this paper, we consider the linearized backward Euler finite element scheme for the Navier-Stokes-Poisson-Nernst-Planck (NSPNP) equations. We aim to derive the unconditional error estimates of all variables in $l^\infty(L^2)$ and $l^\infty(H^1)$ norms. Compared with some previous results, we remove the mesh restriction. In addition, we derive two important physical properties for the proposed numerical scheme. Finally, three numerical examples are implemented to verify the theoretical analysis.

1. Introduction

The NSPNP equations describe the dynamics of electrically charged fluids, the motions of ionized particles or molecules and their interactions with electric fields and the surrounding fluid, which are widely used in colloidal science, biophysics, semiconductors, micro- and nanofluidics, and electrochemistry [6, 4, 7, 2]. The NSPNP equations are given as follows:

$$\begin{cases} u_t - \Delta u + (u\cdot\nabla)u + \nabla p = -(c_1 - c_2)\nabla\psi, & \text{in } \Omega\times(0,T], \\ \mathrm{div} u = 0, & \text{in } \Omega\times(0,T], \\ c_{1t} - \Delta c_1 - \mathrm{div}(c_1\nabla\psi) + (u\cdot\nabla)c_1 = 0, & \text{in } \Omega\times(0,T], \\ c_{2t} - \Delta c_2 + \mathrm{div}(c_2\nabla\psi) + (u\cdot\nabla)c_2 = 0, & \text{in } \Omega\times(0,T], \\ -\Delta\psi = c_1 - c_2, & \text{in } \Omega\times(0,T], \\ u = 0, \frac{\partial c_1}{\partial n} = \frac{\partial c_2}{\partial n} = \frac{\partial \psi}{\partial n} = 0, & \text{on } \partial\Omega\times(0,T], \\ c_1(x,0) = c_{10}(x), c_2(x,0) = c_{20}(x), u(x,0) = u_0(x), \end{cases} \tag{1}$$

where $\Omega \in \mathbb{R}^d, d = 2, 3$ is a bounded domain with a Lipschitz continuous boundary $\partial\Omega$, n is the outward unit-normal vector to $\partial\Omega$, c_1 and c_2 are the concentrations of positive and negative charges, u and p are the velocity and pressure of the fluid, and ψ is the quasi-electrostatic potential.

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Numerous numerical methods have been developed for the PNP and NSPNP equations. It is crucial to preserve some important physical properties at the discrete level, such as the mass conservation, concentration positivity and energy stability [1, 3, 5].

2. Preliminary

We introduce the following function spaces:

$$\begin{cases} X = H^1(\Omega), \quad X_0 = \{\eta \in H^1(\Omega) : \int_\Omega \eta \, dx = 0\}, \\ V = (H_0^1(\Omega))^d, \quad Q = L_0^2(\Omega) = \{q \in L^2(\Omega) : \int_\Omega q \, dx = 0\}. \end{cases} \tag{2}$$

The trilinear forms $b_1(\cdot;\cdot,\cdot)$ on $V \times V \times V$ and $b_2(\cdot;\cdot,\cdot)$ on $V \times X \times X$ are defined by:

$$b_1(u; v, w) = \frac{1}{2}((u \cdot \nabla)v, w) - \frac{1}{2}((u \cdot \nabla)w, v), \quad \forall u, v, w \in V, \tag{3}$$

$$b_2(u; c, \eta) = \frac{1}{2}(u \cdot \nabla c, \eta) - \frac{1}{2}(u \cdot \nabla \eta, c), \quad \forall u \in V, c, \eta \in X. \tag{4}$$

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